

A Triplet Deep Neural Networks Model for Customer Credit Scoring

Jin Xiao*, Runhua Wang

Business School, Sichuan University, Chengdu, Sichuan, 610064, China

*Corresponding author's e-mail: xjxiaojin@126.com

Abstract. Deep neural networks are widely used in speech recognition and face verification with excellent performance, and they are gradually applied and developed in the field of customer credit scoring. Traditional credit scoring work relies on the two-step modeling process of feature processing and model building, which cannot effectively balance data dimensionality and model performance. Based on this, we put forward a triplet deep neural network model for customer credit scoring. This model makes use of the feature that deep neural networks and metric learning can efficiently extract and utilize data feature information so that two samples with the same label are embedded tightly while two samples with different labels are embedded loosely, so as to improve the accuracy of credit scoring. All experiments are conducted on three customer credit scoring datasets. We select accuracy, precision, recall, f1-score and AUC to evaluate the classification performance of all models. The experiments show that the triplet deep neural networks model can perform customer credit scoring more accurately compared with the now commonly used random forest (RF), deep neural networks (DNN), logistic regression (LR), k-nearest neighbor (KNN) and support vector machine (SVM).

Keywords: credit scoring; triplet; deep neural networks

I. Introduction

With the continuous development of financial industry, financial institutions hold hundreds of millions of loans. The credit department has become a huge industry with important economic importance. The large-scale stock loans show that even if the accuracy of the credit scoring model is slightly improved, it can bring accurate and considerable economic benefits[1]. Therefore, one of the main research focuses for financial institutions is how to develop an accurate credit scoring model. Credit scoring is a well-known credit analysis task, which focuses on distinguishing between good and bad loan applicants. The main objective is to evaluate the default probability of the loan applicant before the loan, that is, the probability that the customer fails to repay the loan within a given time limit. Therefore, it is important to construct a classification prediction model to divide each loan applicant, and the customer who is judged to be performing will get the loan, otherwise, it will be rejected.

In the past few decades, many classification technologies have been applied to the field of customer credit scoring. The existing credit scoring models mainly included statistic and machine learning methods[2]. LR and linear discriminant analysis are widely used statistic methods for credit scoring. The application of these statistical methods needs to follow strict mathematical assumptions. However, the characteristics obtained in real life are difficult to meet these strict conditions. The classification effect of machine learning method is

superior to the statistical learning method, and its representative models include DNN, DT (decision trees), KNN, and SVM[3-4].

The existing models have achieved good results, but the accuracy still needs further improvement. The Internet and various smart devices have generated a large amount of credit data with the prosperity of information science, and factors affecting credit have been discovered continuously. While the data dimension is increasing, the amount of data is also expanding, which makes the credit assessment models based on traditional machine learning methods more limited. Therefore, there is a gradual trend toward deep learning algorithms for credit scoring models, and deep learning has been widely researched and achieved great success in many fields. Some of the existing credit scoring models need to rely on a given metric, such as k-means for clustering uses Euclidean distance to calculate distance from sample points to center, and the KNN algorithm also uses the Euclidean distance. Considering that the existing models cannot output large between-class or small within-class variance well, this paper put forward a triplet deep neural networks model for customer credit scoring, which can extract and utilize the feature information of data with greater validity and solve the problem that the traditional two-step modeling process cannot balance data dimensionality and model performance.

II. Related work

A. Triplet network

The triplet network, a kind of deep metric learning algorithm is widely concerned in recent years. It learns effective feature expression by comparing distances. A triplet network is composed of three instances of neural networks sharing parameters. The samples input to these three networks are called anchor, positive and negative samples. Anchor samples and positive samples are in the same category, while negative samples have different category labels. In feature space, samples with the same class behave more closely together and form better classification clusters. Thus, the main objective of the triplet network is to ensure that two samples with the same label are embedded more tightly, while two samples with different labels are embedded more loosely.

Figure 1 is a triplet network structure. During an experiment of neural network training, each triplet (x_a, x_p, x_n) is composed of an anchor sample x_a , a positive sample x_p and a negative sample x_n . The category label corresponding to the triplet satisfies $y_a = y_p \neq y_n$.

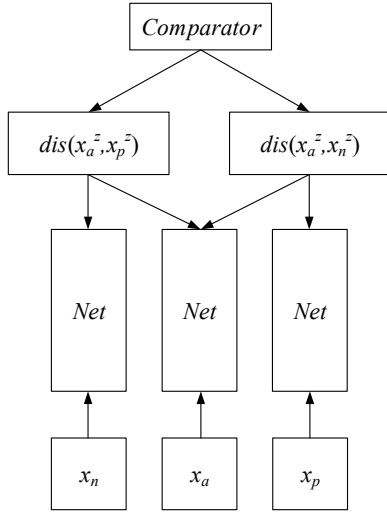


Figure 1. Structure of triplet network.

In Figure 1, $dis(x_a^z, x_p^z)$ represents the Euclidean distance between the anchor class and the positive class, $dis(x_a^z, x_n^z)$ represents the Euclidean distance between the anchor class and the negative class, and the comparator is used to process the distance vector. In the process of triplet network operation, through the $loss(T_z)$ function training can pull in similar samples to separate samples with different labels, and the objective function $loss(T_z)$ can be expressed as[5]:

$$loss(T_z) = \max \left[dis(x_a^z, x_p^z) - dis(x_a^z, x_n^z) + m, 0 \right] \quad (1)$$

The ultimate goal of training is to minimize $dis(x_a^z, x_p^z)$ and maximize $dis(x_a^z, x_n^z)$. That is, $loss(T_z) \rightarrow 0$, the goal is achieved.

B. DNN

DNN is an artificial neural network with feed-forward way. It can be divided into an input, hidden and output layer based on its location of the nodes. It has more hidden layers than shallow networks and can generally contain more neurons. Each variable in DNN has a corresponding relationship. Suppose a DNN has $(N+1)$ layers, Where layer 0 and layer N are the input and output layers respectively, and layers 1 to $(N-1)$ are the hidden layers of the model. Therefore for any n^{th} layer (where $n \in (0, N)$), there is the following corresponding relationship:

$$\begin{cases} net_k^n = \sum_{j=1}^{N^{n-1}} \omega_{j,k}^n \cdot z_j^{n-1} + b_k^n \\ z_k^n = f_l(net_k^n) \end{cases} \quad (2)$$

In the formula, net_k^n is the input value of the k^{th} node in the n^{th} layer, N^{n-1} is the number of nodes in the $(n-1)^{th}$ layer, $\omega_{j,k}^n$ is the weight between the j^{th} node in the n^{th} layer and the

k^{th} node in the n^{th} layer, z_k^{n-1} is the output value of the k^{th} node in the $(n-1)^{th}$ layer, b_k^n is the bias of the k^{th} node in the n^{th} layer, $f_l(\cdot)$ is the activation function of the n^{th} layer.

III. Triplet deep neural networks model

DNN is a neural network with many layers, which can solve complex regression and classification tasks. DNN can automatically conduct comprehensive processing and operation on data to extract useful feature information and then conduct training and prediction, thus effectively reducing human intervention and organically integrating feature processing and model training. The goal of the triplet network structure and its loss function is to make samples having the same label closer others farther apart. If the default class and the performance class can be well distinguished, the classification task will become simpler[6]. This paper combines DNN with triplet network to build a new model suitable for customer credit scoring, so as to more effectively balance the data dimension and model performance, and simplify the classification process.

The customer credit scoring data is usually two-dimensional data, and the image data used for face recognition is usually three-dimensional. Therefore, this paper uses DNN to replace the convolutional neural network commonly used in triplet networks. At the same time, this paper optimizes the sampling mode, and ensures fast convergence by selecting the most appropriate way between offline sampling and online sampling.

When performing offline triplet mining, data processing needs to be done in advance rather than during the triplet network training process. This process requires training the labeled data to form a feature space. This space can be formed by DNN with a softmax loss function, which can also extract the embeddings of last layer. Therefore, data $X1$ is chosen as the training set to form DNN by training labeled data. After training DNN, we train another set of data $X2$ in the feature space (where $X1 \cup X2 = X$ and $X1 \cap X2 = \emptyset$). A distance matrix is placed in the feature space, and we can find the triplet samples satisfying the condition by using the distance matrix[7]. After the sample pairs in the form of triples were prepared, the triplet deep neural networks (triplet DNN) model is used for training.

When performing online triplet mining, data is processed in small batches in the training stage, that is, triplets are in small batch data, rather than being input in the form of triplet DNN. For the customer credit scoring problem, randomly sample P classes to form a batch, and then randomly sample K samples of every class to obtain a batch of PK samples. For each sample anchor in the batch, when forming triplets, you can select the more difficult samples in the batch to calculate the loss, which is called batch semi-hard. For this sample selection method, the optimization objective is:

$$L_{BSH} = \sum_{i=1}^P \sum_{a=1}^K \sum_{\substack{p=1 \\ p \neq a}}^K \left[m + D(y_a^i, y_p^i) - \left\{ D(y_a^i, y_n^j) \mid D(y_a^i, y_n^j) > D(y_a^i, y_p^i), \right. \right. \\ \left. \left. j = 1 \dots P, n = 1 \dots K \right\} \right]_+ \quad (3)$$

Another online triplet mining method is to simply use all possible combinations of $PK(PK-K)(K-1)$, which is called the batch all method. The optimization objective of this method is:

$$\begin{cases} L_{BA}(\theta; X) = \sum_{i=1}^P \sum_{a=1}^K \sum_{\substack{p=1 \\ p \neq a}}^K \sum_{\substack{j=1 \\ j \neq i}}^P \sum_{n=1}^K \left[m + d_{j,a,n}^{i,a,p} \right]_+ \\ d_{j,a,n}^{i,a,p} = D(f_\theta(x_a^i), f_\theta(x_p^i)) - D(f_\theta(x_a^i), f_\theta(x_n^j)) \end{cases} \quad (4)$$

In the process of the experiment, how to choose triplet mining methods depends on the distribution of specific datasets. This paper chooses the triplet mining method by the results of cross-validation.

Under the above conditions, assuming that the original dataset of customer credit is D , the steps of the triplet DNN model for credit scoring are as follows:

Step 1: Preprocess the data and divide it into training set D_{train} and testing set D_{test} at a ratio of 7:3.

Step 2: Divide D_{train} into n equal parts, and construct several triplet sample pairs (anchor, positive, negative) by online and offline training methods based on formulas (1), (3) and (4).

Step 3: Use n -fold cross-validation to train the triplet DNN model. If the loss function is no longer reduced, the training is stopped. The results of cross-validation are used to adjust the model's hyperparameters and select the final sample pair construction method.

Step 4: Classify D_{train} and D_{test} respectively to obtain the classification results $R=(R_1, R_2, \dots, R_N)$ and $r=(r_1, r_2, \dots, r_N)$.

IV. Empirical analysis

A. Data

This paper selects three public credit-scoring datasets to carry out the experiment. The dataset German with 1000 samples and 20 features is derived from the UCI database. The dataset home credit blending (Home) is from a finance company, including 30751 samples and 718 characteristics. The dataset gives me some credit (GiveCre) and is provided by a financial institution for the Kaggle competition, including 150000 samples and 10 features. All datasets were pre-processed according to the following steps: (1) delete meaningless feature variables, such as ID and loan number; (2) missing value processing: delete characteristic variables with a missing rate of more than 60%, and interpolate the missing values of numerical variables and nominal variables with mean and mode respectively; (3) numerical conversion: the nominal variable is converted into one hot and the Z-score method is used to standardize the data.

B. Evaluation measures

In view of the unbalanced categories of customer credit scoring data, the traditional accuracy indicators cannot reflect the prediction performance of this model on the positive category. According to the benchmark research recommendations of Lessmann et al.[8], future research should use at least three performance indicators to evaluate credit scoring models. In this paper, we use accuracy, precision, recall, f1-score and AUC to evaluate the classification effect.

For customer credit assessment, each customer sample can be divided into performance and default categories. Generally, a few categories are marked as positive and most categories are marked as negative. In the case of customer credit assessment, there are usually fewer default categories, so the default category is marked as positive. As shown in **Table 1**, to calculate the evaluation measures, we need to define a second-order confusion matrix.

Table 1. The definition of the confusion matrix.

Predicted Labeled	Positive	Negative
	True Positive (TP)	False Negative (FN)
Positive		
Negative	False Positive (FP)	True Negative (TN)

Accuracy represents the amount of the data correctly classified as a percentage of the entire:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

Precision is the ratio of correctly classified positive samples to samples predicted to be positive:

$$Precision = \frac{TP}{TP + FP} \quad (6)$$

Recall represents the proportion of the number of positive classes with correct model classification to the number of positive classes with real data:

$$Recall = \frac{TP}{TP + FN} \quad (7)$$

F1-score is a comprehensive index that is regarded as the harmonic mean of recall and precision:

$$F1-score = \frac{2 \cdot Recall \cdot Precision}{Recall + Precision} \quad (8)$$

AUC is used to calculate the area enclosed by the ROC curve with the x-axis, where the ROC curve is plotted with the FP rate and the TP rate as the horizontal and vertical axes respectively. AUC is hardly affected by the unbalanced data:

$$AUC = \int_0^1 f_{ROC}(x) dx \quad (9)$$

C. Empirical results

In this paper, three customer credit scoring datasets described in Section 4.1 are used for experiments, and five

indicators described in Section 4.2 are selected to evaluate the classification performance. Triplet DNN is compared with RF, DNN, LR, KNN and SVM are often used in customer credit scoring. The experimental results are shown in **Table 2**.

Table 2. Experimental results (%).

Measures	Datasets	RF	DNN	LR	KNN	SVM	Triplet DNN
Accuracy	German	75.67 (4)	74.33 (5)	76.33 (3)	71.20 (6)	77.00 (1)	76.67 (2)
	Home	91.77 (1)	91.23 (5)	91.40 (4)	91.74 (2)	89.91 (6)	91.56 (3)
	GiveCre	93.50 (6)	96.22 (1)	93.63 (4)	93.64 (3)	93.59 (5)	94.73 (2)
Precision	German	65.45 (3)	62.26 (4)	66.10 (2)	53.13 (6)	68.42 (1)	60.41 (5)
	Home	1.00 (6)	30.77 (3)	37.52 (2)	4.44 (5)	24.01 (4)	38.46 (1)
	GiveCre	52.83 (4)	75.00 (1)	57.42 (3)	61.29 (2)	48.50 (6)	50.00 (5)
Recall	German	40.00 (5)	36.67 (6)	43.33 (3)	44.77 (2)	43.33 (3)	64.44 (1)
	Home	0.13 (5)	8.16 (4)	11.50 (1)	0.05 (6)	10.39 (2)	10.20 (3)
	GiveCre	16.41 (3)	27.27 (2)	14.89 (4)	10.75 (5)	5.02 (6)	51.85 (1)
F1-score	German	49.66 (4)	46.15 (5)	52.35 (3)	39.11 (6)	53.06 (2)	57.45 (1)
	Home	0.26 (5)	12.90 (3)	6.79 (4)	0.03 (6)	14.51 (2)	16.13 (1)
	GiveCre	25.04 (3)	40.00 (2)	23.65 (4)	18.29 (5)	8.97 (6)	50.91 (1)
AUC	German	73.32 (5)	74.41 (4)	76.10 (2)	70.42 (6)	76.04 (3)	79.94 (1)
	Home	70.32 (4)	72.13 (3)	73.29 (1)	58.66 (6)	67.02 (5)	73.03 (2)
	GiveCre	81.93 (3)	82.54 (2)	81.73 (4)	80.50 (5)	78.96 (6)	84.74 (1)
Mean		49.15 (4.07)	55.34 (3.33)	53.74 (2.93)	46.53 (4.73)	50.58 (3.87)	60.03 (2.00)

The table lists the classification performance of the triplet DNN model and other common credit scoring models constructed in this article on the three datasets. The specific values of each scoring index are listed in the table. The numbers in brackets indicate the ranking order of specific datasets on specific indicators using different models. The last row in the table indicates the overall average value and an average ranking of each model on all indicators. It is found that the overall average value and an average ranking of the evaluation indexes of the triplet DNN model presented in this article are the best, which are 60.03% and 2.00 respectively. The overall average value and an average ranking of DNN and LR are better than SVM, RF and KNN. The overall average value and average ranking of the KNN model are the lowest among all models. Its classification performance on the three datasets is poor.

V. Conclusion

This article proposes a triplet DNN model for customer credit scoring, which can help optimize the mapping and embedding, and better distinguish the samples located at the classification boundary. From the experimental results, we can see that there are some differences in the effects of the same models on different datasets, which may be due to their different data distributions, resulting in their different suitable fitting methods. However, the triplet DNN model described in this article has better classification results in customer credit scoring.

Acknowledgement

This work was supported by the National Natural Science Foundation of China (72171160).

References

- [1] Xia H S, Wang Y, Gauthier J, et al. Knowledge graph of mobile payment platforms based on deep learning: Risk analysis and policy implications[J]. *Expert Systems with Applications*, 2022, 208: 118143.
- [2] Li Z, Zhang J, Yao X, et al. How to identify early defaults in online lending: A cost-sensitive multi-layer learning framework[J]. *Knowledge-Based Systems*, 2021, 221: 106963.
- [3] He H, Zhang W, Zhang S. A novel ensemble method for credit scoring: Adaption of different imbalance ratios[J]. *Expert Systems with Applications*, 2018, 98: 105-117.
- [4] Gomez J A, Arevalo J, Paredes R, et al. End-to-end neural network architecture for fraud scoring in card payments[J]. *Pattern Recognition Letters*, 2018, 105: 175-181.
- [5] Shi D, Orouskhani M, Orouskhani Y. A conditional Triplet loss for few-shot learning and its application to image co-segmentation[J]. *Neural Networks*, 2021, 137: 54-62.
- [6] He G, Zhang Q, Zhang H, et al. A concept ontology triplet network for learning discriminative representations of fine-grained classes[J]. *Multimedia Tools and Applications*, 2020, 79(33): 25189-25214.
- [7] Sikaroudi M, Ghogh B, Safarpour A, et al. Offline Versus Online Triplet Mining Based on Extreme Distances of Histopathology Patches[C]. *Cham*, 2020, Springer International Publishing, 333-345.
- [8] Lessmann S, Baesens B, Seow H-V, et al. Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research[J]. *European Journal of Operational Research*, 2015, 247(1): 124-136.